Gerstner (2000) — Population Dynamics of Spiking Neurons: Fast Transients, Asynchronous States, and Locking

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A reading of Gerstner (2000), *Population Dynamics of Spiking Neurons: Fast Transients, Asynchronous States, and Locking* (Neural Computation 12(1):43–89, communicated by Carl van Vreeswijk). It is the population/mean-field companion to the microscopic balanced state we build in nb058 and the gamma rhythm in nb050: the same asynchronous-vs-rhythmic question, viewed from above the spikes.

In plain English. Stop tracking individual neurons and track a single number — the fraction of the population firing right now, $A(t)$. Gerstner writes one equation for how $A(t)$ evolves, then asks two things of it. *When does the population fire steadily and independently?* (asynchronous). *When does it collapse into a synchronous pulse train?* (locking). The answer is a phase diagram in two knobs: how noisy the neurons are, and how long their signals take to arrive. High noise keeps the population asynchronous; low noise (or short delays) tips it into a rhythm. That boundary is precisely the AI-to-gamma transition our notebooks explore — derived here analytically rather than simulated.

From neurons to a population

In a pool of N statistically identical neurons, count the spikes $n_{\mathrm{act}(t;t+\Delta t)}$ in a short window and divide by the window and the population:

$$A(t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \frac{n_{\mathrm{act}(t;t+\Delta t)}}{N}. \quad (1)$$

where:

- $A(t)$ — the **population activity**, a firing rate per neuron (units of kHz here);
- N — number of neurons in the homogeneous pool;
- n_{act} — spikes emitted by the pool in $[t, t+\Delta t)$.

This is the only observable we track. The trick that makes it self-contained is *renewal*: an integrate-and-fire (or spike-response) neuron with no adaptation is reset at every spike and forgets its past, so its future depends only on the time $\hat{t}$ of its **last** spike and the input it has seen since. That short memory is what closes the equation below.

The population activity equation

Every neuron firing at time t last fired at some earlier $\hat{t}$. Group neurons by their last firing time and the equation writes itself:

$$A(t) = \int_{-\infty}^{t} P_h\big(t \mid \hat{t}\big) A\big(\hat{t}\big)\, \mathrm{d}\hat{t}, \quad (2)$$

where:

- $P_h\big(t \mid \hat{t}\big)$ — the **interspike-interval density**: the probability per unit time that a neuron which last fired at $\hat{t}$ fires its next spike at t, given the input it received in between. It is normalised, $\int_{\hat{t}}^{\infty} P_h\big(t \mid \hat{t}\big)\, \mathrm{d}t = 1$ (every neuron fires again);
- the subscript h — the dependence on the **input potential** $h(t)$, the membrane drive from synapses and external sources.

Read it plainly: the activity *now* is the activity at every past instant $\hat{t}$, weighted by how likely those neurons are to fire their next spike exactly now. The recurrent coupling enters through h: in the mean field each neuron's drive is the population activity filtered by the synapse,

$$h(t) = J_0 \int_0^\infty \varepsilon(s) A(t-s)\,\mathrm{d}s, \quad (3)$$

with J_0 the total recurrent coupling strength (sum of synaptic weights, $O(1)$) and $\varepsilon(s)$ the postsynaptic response kernel — the shape of one synaptic event. The activity drives the potential; the potential reshapes the firing density P_h; that feeds back into the activity. Gerstner shows the classic Wilson–Cowan rate equation is a coarse special case of this.

Locking: when a rhythm sustains itself

Suppose the pool is already nearly synchronous — it fires in tight pulses of width 2δ and period T. Does that pulse train continue, and does it *sharpen* or smear out? Gerstner's answer (the Gerstner–Ritz–van Hemmen (1996) **locking theorem**, recovered here from the population equation) is a one-line criterion:

> A synchronous pulse self-stabilises when neurons fire on the **rising** flank of their input potential, $h'(t) > 0$ at the firing time.

The mechanism is a compression of firing-time differences. If h is rising as the pulse arrives, a neuron that was slightly late sees a slightly larger drive and catches up; the spread of firing times shrinks, the pulse narrows, and the activity peak grows. Formally the compression is equivalent to a growing activity, $h'(t) > 0 \iff A(t) > A(t-T)$. Fire on the rising edge and synchrony feeds itself; fire on the falling edge and the pulse disperses.

This is exactly the PING loop of nb050 stated at the population level: inhibition recruited on the rising edge of recurrent excitation tightens each gamma volley. The locking theorem is *why* a gamma rhythm is a stable attractor of this hardware, not a coincidence of one parameter set.

Asynchronous firing: a fixed point, and when it survives

The opposite regime is **incoherent** firing: every neuron fires at the same mean rate but at random relative phases, so the population activity is flat,

$$A(t) = A_0 = \frac{1}{\langle T \rangle}, \quad (4)$$

the inverse mean interspike interval. This constant is always a solution of the population equation — the question is whether it is *stable*. Perturb it with a small oscillation and ask whether the oscillation grows or decays:

$$A(t) = A_0 + \hat{A}_1 e^{\lambda t + i\omega t}, \quad \hat{A}_1 \ll A_0. \quad (5)$$

where ω is the candidate oscillation frequency and λ its growth rate. Substituting equation (5) into the linearised form of the population equation (2) and cancelling the common factor gives the **bifurcation condition** — the heart of the paper:

$$1 - \hat{P}(\omega) = i\omega J_0 A_0 \hat{\varepsilon}(\omega) \hat{L}(\omega), \quad (6)$$

where:

- $\hat{P}(\omega)$ — Fourier transform of the interspike-interval density (how the single neuron's firing statistics filter a perturbation);
- $\hat{\varepsilon}(\omega)$ — Fourier transform of the synaptic kernel (the synapse's frequency response);
- $\hat{L}(\omega)$ — the neuron's **linear response filter**: how strongly a small oscillation in the input current modulates its firing.

Solutions of equation (6) with $\lambda = 0$ are the marginal points. Inside the curve the incoherent state is stable ($\lambda < 0$, perturbations decay); cross it and $\lambda > 0$ — the flat activity breaks into a population oscillation at frequency ω.

The payoff is the **phase diagram** in two physical knobs — the neuronal **noise** level σ and the synaptic/axonal **delay** Δ:

- **High noise -> asynchrony is stable.** Jitter in the single-neuron firing times ($\hat{P}$ broad) damps any coherent perturbation; the pool stays incoherent at A_0.
- **Low noise -> asynchrony loses stability** and the population oscillates, at ω near the firing rate $2\pi/T_0$ or its harmonics (neurons fire in groups several times faster than their individual rate).
- **Delay selects the unstable frequency.** Reducing Δ can itself trigger the instability at $\omega_1 = 2\pi/T_0$; longer delays favour higher harmonics.

So the AI-to-rhythm transition is, in this theory, a Hopf bifurcation of the asynchronous fixed point, with noise and delay as the control parameters.

The bridge to the balanced state — where the noise comes from

Here is the tension worth dwelling on, and the reason this paper pairs so cleanly with nb058. **For Gerstner, the noise level σ is an input parameter** — you dial it, and it tells you whether asynchrony is stable. He is explicit (§8.2) that fluctuations arising from the *internal recurrent coupling* are hard to treat exactly, and points to Brunel & Hakim (1999) for the sparse case.

That gap is exactly what Vreeswijk–Sompolinsky — and our nb058 — fill from below. In the strongly coupled balanced state the noise is **not external**: the recurrent dynamics are deterministically chaotic, and the aperiodic spike traffic each cell receives *is* the noise. So the two pictures compose:

- **V&S / nb058** explains *where* the noise comes from: self-generated chaos (we measured a positive Lyapunov exponent on frozen input).
- **Gerstner** explains, *given* that noise, whether the population sits in the stable-asynchronous corner or tips into a rhythm.

A balanced network manufacturing its own high effective σ is precisely what parks it in Gerstner's "asynchrony-stable" region without any external noise source — and lowering that self-generated noise (weaker coupling, more shared input) is what walks it across the bifurcation into the gamma regime of nb050.

Why the asynchronous state is useful

Gerstner closes (§8.2) with the functional point. The incoherent state at low noise is not just a resting condition — it transmits signals *fast*. Because the population can respond to a change in input current without first dephasing a rhythm, the activity $A(t)$ follows input transients with essentially no high-frequency cutoff (for his analytically clean noise model). The asynchronous state is a low-latency communication channel; the rhythmic state is a clock. The same network, on either side of one bifurcation, offers both — which is the thread running through this whole collection.

One-paragraph summary

Track the firing fraction $A(t)$; it obeys a renewal integral equation closed by the input potential h. A flat A_0 (asynchrony) and a pulsed $A(t)$ (locking) are both solutions. Locking is stable when neurons fire on the rising edge of h (the PING mechanism); asynchrony is stable when the neurons are noisy enough, and loses stability — via a Hopf bifurcation set by noise and delay — into a population rhythm. The one ingredient the theory takes as given, the noise, is the very thing the balanced-state chaos of nb058 supplies. Two lenses, one boundary.